

Oracle Cloud, AWS Fundamentals & Cloud Architecture Mastery Syllabus

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Program Overview

This syllabus is designed to build complete expertise in:

- Cloud Computing Fundamentals
- Oracle Cloud Infrastructure (OCI)
- Amazon Web Services (AWS)
- Cloud Architecture
- Infrastructure Design
- Security & Networking
- DevOps on Cloud
- Kubernetes & Containers
- High Availability & Scalability
- Cloud Automation
- Monitoring & Optimization
- Enterprise Cloud Projects

This syllabus is suitable for:

- Beginners
 - Cloud Engineers
 - DevOps Engineers
 - Solution Architects
 - System Administrators
 - IT Trainers
 - Infrastructure Engineers
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Phase 1 — Fundamentals of Cloud Computing

Module 1: Introduction to Cloud Computing

Topics

- What is Cloud Computing?
- Evolution of Cloud Computing

- Traditional vs Cloud Infrastructure
- Cloud Service Models
- IaaS
- PaaS
- SaaS
- Cloud Deployment Models
- Public Cloud
- Private Cloud
- Hybrid Cloud
- Multi Cloud
- Benefits of Cloud Computing
- Challenges in Cloud
- Shared Responsibility Model

Hands-on

- Understanding cloud platforms
 - Exploring cloud dashboards
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Module 2: Core Cloud Concepts

Topics

- Virtualization
- Hypervisors
- Containers
- Microservices
- Scalability
- Elasticity
- High Availability
- Fault Tolerance
- Disaster Recovery
- Load Balancing
- CDN Concepts
- Edge Computing

Hands-on

- Virtual machine deployment
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Phase 2 — Networking Fundamentals for Cloud

Module 3: Networking Basics

Topics

- OSI Model
- TCP/IP Model
- IP Addressing
- CIDR
- Subnetting
- DNS
- DHCP
- NAT
- VPN
- Firewalls
- Routing
- Switching
- Proxies
- Ports & Protocols

Hands-on

- IP subnet calculations
 - Network troubleshooting
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Module 4: Cloud Networking Architecture

Topics

- VPC Concepts
- Virtual Cloud Networks
- Public & Private Subnets
- Route Tables
- Internet Gateways
- NAT Gateways
- Security Lists
- Network Security Groups
- Peering
- Transit Routing
- Hybrid Connectivity

Hands-on

- Designing enterprise cloud networks

Phase 3 — AWS Fundamentals

Module 5: Introduction to AWS

Topics

- AWS Global Infrastructure
- Regions & Availability Zones
- AWS Console
- AWS CLI
- AWS Free Tier
- IAM Basics

Hands-on

- AWS account setup
 - CLI configuration
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Module 6: AWS Compute Services

Topics

- EC2
- AMIs
- Auto Scaling
- Elastic Load Balancer
- Lambda
- ECS
- EKS
- Elastic Beanstalk

Hands-on

- Launching EC2 instances
 - Auto Scaling setup
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Module 7: AWS Storage Services

Topics

- S3
- EBS
- EFS

- Glacier
- Storage Classes
- Lifecycle Policies
- Backup Strategies

Hands-on

- Static website hosting on S3
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Module 8: AWS Database Services

Topics

- RDS
- DynamoDB
- Aurora
- Redshift
- ElastiCache
- Database Migration Service

Hands-on

- Deploying MySQL on RDS
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Module 9: AWS Networking Services

Topics

- VPC
- Route Tables
- Internet Gateway
- NAT Gateway
- Security Groups
- NACLs
- Route53
- CloudFront
- Direct Connect

Hands-on

- Multi-tier VPC architecture
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Module 10: AWS Security & IAM

Topics

- IAM Users & Roles
- Policies
- MFA
- KMS
- Secrets Manager
- AWS Shield
- AWS WAF
- Security Best Practices

Hands-on

- IAM role creation
 - Secure architecture implementation
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Module 11: AWS Monitoring & Management

Topics

- CloudWatch
- CloudTrail
- Config
- Trusted Advisor
- Systems Manager
- Cost Explorer

Hands-on

- Monitoring EC2 instances
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Phase 4 — Oracle Cloud Infrastructure (OCI)

Module 12: Introduction to OCI

Topics

- OCI Overview
- OCI Global Infrastructure
- Compartments
- OCI Console
- OCI CLI

- OCI Identity & Access Management

Hands-on

- OCI account setup
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Module 13: OCI Compute Services

Topics

- OCI Compute Instances
- Shapes
- Custom Images
- Autoscaling
- Instance Pools
- Functions
- OKE (Oracle Kubernetes Engine)

Hands-on

- Deploying Linux VMs on OCI
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Module 14: OCI Storage Services

Topics

- Object Storage
- Block Volumes
- File Storage
- Archive Storage
- Backup Policies

Hands-on

- Object storage configuration
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Module 15: OCI Networking

Topics

- VCN
- Subnets
- Route Tables
- Internet Gateway

- NAT Gateway
- Service Gateway
- Security Lists
- NSGs
- FastConnect
- DRG

Hands-on

- OCI enterprise networking setup
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Module 16: OCI Databases

Topics

- Autonomous Database
- Oracle DB Systems
- Exadata Basics
- NoSQL Database
- Data Safe

Hands-on

- Deploying Autonomous Database
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Module 17: OCI Security & Monitoring

Topics

- IAM Policies
- Vault
- Security Zones
- Cloud Guard
- Logging
- Monitoring
- Notifications

Hands-on

- Implementing OCI security best practices
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Phase 5 — Cloud Architecture

Module 18: Cloud Architecture Fundamentals

Topics

- Cloud Design Principles
- Well-Architected Framework
- Scalability Design
- Resilient Architectures
- High Availability Design
- Fault Tolerance
- Performance Optimization

Hands-on

- Designing scalable systems
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Module 19: Enterprise Cloud Architecture

Topics

- 3-Tier Architecture
- Microservices Architecture
- Serverless Architecture
- Event-driven Architecture
- Multi-region Architecture
- Hybrid Cloud Architecture
- Multi-cloud Architecture

Hands-on

- Enterprise architecture case studies
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Module 20: Security Architecture

Topics

- Zero Trust Security
- Encryption at Rest & Transit
- Identity Federation
- SIEM Concepts
- SOC Integration
- Security Compliance

Standards

- ISO 27001
 - GDPR
 - HIPAA
 - PCI-DSS
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Phase 6 — DevOps on Cloud

Module 21: DevOps Fundamentals

Topics

- CI/CD Basics
- Git & GitHub
- Jenkins
- GitHub Actions
- Infrastructure as Code
- Terraform
- Ansible

Hands-on

- CI/CD pipeline creation
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Module 22: Containers & Kubernetes

Topics

- Docker Basics
- Docker Compose
- Kubernetes Architecture
- Pods
- Deployments
- Services
- Helm
- Ingress
- OKE
- EKS

Hands-on

- Kubernetes application deployment
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Phase 7 — Cloud Automation

Module 23: Infrastructure as Code

Topics

- Terraform Basics
- Terraform State
- Modules
- Variables
- OCI Terraform Provider
- AWS Terraform Provider
- CloudFormation Basics

Hands-on

- Automating cloud infrastructure
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Module 24: Scripting & Automation

Topics

- Bash Scripting
- Python Automation
- AWS Boto3
- OCI SDK
- Automation Workflows

Hands-on

- Automated provisioning scripts
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Phase 8 — Monitoring, Logging & Cost Optimization

Module 25: Monitoring & Logging

Topics

- Centralized Logging
- Monitoring Dashboards
- Alerts & Notifications

- ELK Stack
- Prometheus
- Grafana

Hands-on

- Real-time monitoring dashboards
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Module 26: Cloud Cost Optimization

Topics

- Cost Management
- Reserved Instances
- Spot Instances
- Resource Optimization
- FinOps Basics
- Budget Monitoring

Hands-on

- Cloud cost analysis
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Phase 9 — Disaster Recovery & Backup

Module 27: DR & Business Continuity

Topics

- Backup Strategies
- Disaster Recovery Planning
- RTO & RPO
- Multi-region Deployment
- Failover Mechanisms
- Backup Automation

Hands-on

- DR architecture implementation
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Phase 10 — Real-Time Cloud Projects

Beginner Projects

- Static Website Hosting
- Cloud Storage Setup
- Linux VM Deployment
- IAM Configuration

Intermediate Projects

- Multi-tier Web Application
- Kubernetes Deployment
- CI/CD Pipeline
- Cloud Monitoring Dashboard

Advanced Projects

- Multi-cloud Enterprise Architecture
 - High Availability Kubernetes Platform
 - AI-powered Cloud Monitoring System
 - Automated DevOps Infrastructure
 - Enterprise Security Monitoring Platform
 - Cloud Cost Optimization Dashboard
 - Hybrid Cloud Infrastructure
 - Disaster Recovery Automation System
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Phase 11 — Interview Preparation

Module 28: Cloud Interview Preparation

Topics

- AWS Interview Questions
- OCI Interview Questions
- Networking Questions
- Kubernetes Questions
- DevOps Questions
- Architecture Design Questions
- Troubleshooting Scenarios

Hands-on

- Mock interviews

- Architecture discussions
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Phase 12 — Certifications

AWS Certifications

- AWS Cloud Practitioner
- AWS Solutions Architect Associate
- AWS DevOps Engineer
- AWS SysOps Administrator

Oracle Certifications

- OCI Foundations Associate
 - OCI Architect Associate
 - OCI DevOps Professional
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Recommended Tools

Cloud Platforms

- AWS
- Oracle Cloud Infrastructure

DevOps Tools

- Docker
- Kubernetes
- Jenkins
- Terraform
- Ansible

Monitoring Tools

- Grafana
- Prometheus
- ELK Stack

Programming

- Python
- Bash

Recommended Learning Timeline

Beginner Level (1–2 Months)

- Cloud Fundamentals
- Networking
- AWS Basics
- OCI Basics

Intermediate Level (3–5 Months)

- Architecture
- DevOps
- Kubernetes
- Security

Advanced Level (6–12 Months)

- Enterprise Cloud Architecture
- Automation
- Multi-cloud Design
- Disaster Recovery
- Monitoring & Optimization

Outcome After Completion

After completing this syllabus, learners can become:

- Cloud Engineer
 - AWS Engineer
 - OCI Engineer
 - DevOps Engineer
 - Site Reliability Engineer
 - Cloud Architect
 - Infrastructure Engineer
 - Platform Engineer
 - Kubernetes Administrator
 - Solutions Architect
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Best Learning Strategy

Daily Routine

- 2 Hours Theory
- 3 Hours Hands-on Labs
- 1 Hour Troubleshooting Practice
- 1 Hour Architecture Design

Weekly Routine

- Deploy one cloud project
 - Practice Terraform automation
 - Build Kubernetes workloads
 - Optimize cloud infrastructure
 - Create architecture diagrams
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Final Capstone Projects

Enterprise-Level Projects

- Multi-cloud Kubernetes Platform
 - Enterprise CI/CD Infrastructure
 - AI-powered Cloud Monitoring System
 - Hybrid Cloud Architecture
 - Intelligent Auto-scaling Platform
 - Cloud-native DevOps Platform
 - Enterprise Security Compliance System
 - High Availability SaaS Architecture
 - Cloud Cost Optimization AI Tool
 - Real-time Distributed Monitoring System
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Final Advice

To become successful in Cloud & Architecture:

1. Master networking deeply
2. Learn Linux administration
3. Practice cloud deployments daily
4. Understand architecture principles
5. Learn Infrastructure as Code

6. Build real-world projects
7. Practice troubleshooting
8. Learn security best practices
9. Master Kubernetes & Containers
10. Stay updated with cloud innovations